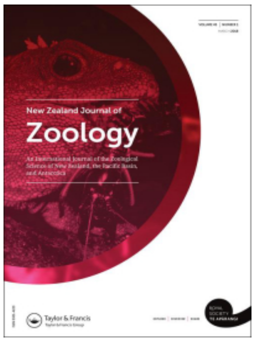

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Possible pathogens of social wasps (Hymenoptera: Vespidae) and their potential as biological control agents

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Possible pathogens of social wasps (Hymenoptera: Vespidae) and their potential as biological control agents

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Abstract No practical methods are currently available for area-wide, long-term control of social wasps in New Zealand. Pathogens have received little attention as potential control agents. Records from wasps of the genera *Vespula*, *Vespa*, and *Dolichovespula* and their associated nest material include 50 fungal, 12 bacterial, 5–7 nematode, 4 protozoan, and 2 viral species, although few have been confirmed through bioassay as pathogens of these wasp species. Despite few naturally-occurring host-specific pathogens and records of diseased colonies, wasps are susceptible to generalist insect diseases in bioassays. Fungi belonging to the genera *Aspergillus*, *Paecilomyces*, *Metarhizium*, and *Beauveria* have been confirmed through bioassay as Vespiniae pathogens, as have the bacteria *Serratia marcescens* and *Bacillus thuringiensis*, and nematodes *Heterorhabditis bacteriophora*, *Steinernema* (= *Neoaplectana*) sp., *S. feltiae*, *S. carpocapsae* and *Pheromermis vesparum*. Several of the pathogens listed here provide a resource from which inundative control agents might be developed, but none have potential as classical self sustaining control agents that can be transferred from generation to generation. As few studies have systematically searched for pathogens, it is likely other candidates suitable for use as control agents may be found.

Keywords Vespinae; *Vespula*; *Vespa*; *Dolichovespula*; social wasps; pathogen; biological control

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INTRODUCTION

Social wasp species from the genera *Vespula*, *Vespa*, and *Dolichovespula* are considered pests in temperate and tropical regions of the world (Smithers & Holloway 1978; Gambino et al. 1987; Stein & Wrensch 1988; Clapperton et al. 1989; Toft & Rees 1998). In New Zealand, *Vespula germanica* was accidentally introduced in the 1940s (Thomas 1960), and *V. vulgaris* in the 1980s (Donovan 1983). Both species are now widespread. Nest densities of *V. vulgaris* in native beech forests are significantly higher than densities in other countries (Thomas et al. 1990; Donovan 1993). Methods for controlling social wasps typically involve luring wasp foragers to traps (Davis et al. 1973; Edwards 1977), the use of insecticides in bait formulation, or direct application of insecticide into nests (Edwards 1980; Spurr 1993). These methods are considered impractical where wasps are causing problems in forests and farmland, as the nests are too numerous to deal with individually, the area is often too large, and/or the terrain inaccessible (Beggs et al. 1998). At best, chemical control achieves only localised and temporary nuisance abatement. In addition, insecticide use is expensive, labour-intensive, potentially hazardous to non-target organisms, and effective only late in the season when there are large numbers of foragers collecting baits. Clearly, further improvements to control options for wasps are needed.

Biological control of wasps has been attempted with limited success. In New Zealand, the parasitoid *Sphecophaga vesparum*, introduced from its native range in Northern Hemisphere temperate regions, has established in some areas (Donovan et al. 1989). Although mathematical models predict the parasitoid may eventually reduce wasp nest density by up to 10% (Barlow et al. 1996), greater levels of control are needed to reduce the detrimental impacts of wasps on native ecosystems (Toft & Rees 1998; Beggs et al. 1998). Other insects known to attack wasp colonies are not considered to be suitable as biological control agents in New Zealand (Donovan 1993).

Pathogens (including entomopathogenic fungi, bacteria, viruses, protozoa, and nematodes) may have the potential to restrict the growth and productivity of wasp colonies (Akre 1991), but have received little study. A pathogen that could overcome colony defence mechanisms and spread within a nest to kill occupants could act as an "inundative" biological control agent, providing short-term population reduction in the treated area and enhance or replace existing toxic baits. For long-term control, the agent would need to survive over winter after wasp nests have died off and be able to disperse to new nests in the spring. Such a pathogen would have potential as a self-sustaining "classical" biological control agent. Pathogens are being used as inundative biological control agents against other social insects, such as fire ants (Siebeneicher et al. 1992; Pereira et al. 1993; Briano et al. 1995), carpenter ants (Kelley-Tunis et al. 1995), and termites (Khan et al. 1993; Zoberi 1995; Milner & Staples 1996), and they have been shown to be effective as inundative controls despite the challenge of controlling insects that have grooming and/or house-cleaning habits.

This review summarises the known pathology of three eusocial genera in the subfamily Vespinae (*Vespula*, *Vespa*, and *Dolichovespula*) and aims to classify the potential of recorded pathogens as biological control agents of social wasps.

POTENTIAL PATHOGENS

Fungi

Records of fungi on Vespinae date back to 1749, when the ecclesiastical naturalist, Torrubia, recorded a wasp infected by "trees" in Cuba (Samson et al. 1988). Although there are relatively few records of fungi infecting wasps themselves, a number of fungi have been recorded from nest material (any material within a nest, including nest paper, bits of dead wasps and old food scraps) (Table 1). Of the fungi listed, only *Aspergillus flavus* and *Beauveria bassiana* have been confirmed pathogenic by Koch's postulates (Table 2). Strains of *Metarhizium anisopliae*, a pathogenic species known to infect insects in many orders, have also been confirmed pathogenic against *Vespula* sp. in bioassays, although it has never been recorded as a naturally occurring pathogen of wasps (Table 1). The strains tested were originally isolated from other insects or soil (T. Glare, unpubl.).

Of the other fungi listed in Table 1, *Aspergillus* spp., *Cladosporium* spp., *Paecilomyces farinosus*, and *Penicillium* spp. only occasionally act as pathogens of insects. More often these fungi act as saprophytes, and may have been associated with nest material or have infected wasps after death. Species of *Alternaria* are usually plant pathogenic or saprophytic, but we recently isolated an *Alternaria* sp. from a diseased *V. vulgaris* cadaver (Glare, Rose & Ratnieks, unpubl. data), and Durrell (1965) recorded an *Alternaria* sp. from nest material. *Cordyceps* is an Ascomycete genus containing a number of species pathogenic to insects and mites. These species would be expected to be true insect pathogens, as they have been found killing wasps, or are commonly recorded killing other insects. As the sexual and asexual stages of fungi are often not produced together (in some cases the sexual stage is never produced) a single fungal species can be classified in both the Deuteromycetes (for asexual fungi) and in another class, such as Ascomycetes. Some asexual linkages for the sexually reproducing *Cordyceps* spp. are known, or predicted. One such link is *Cordyceps humberti* and *Hirsutella saussurei* (Kobayasi 1941). *Hirsutella saussurei*, probably the fungus recorded by Torrubia in 1749, is an occasional pathogen of adult *Vespula* spp. in New Zealand (Glare & Harris, unpubl. data), and has been recorded in several other countries (Table 1). It is always reported on adults found away from the colony rather than in the nest. Although *Botrytis cinerea* has been found in nests (Acolat 1953; Cymorek 1978), it is a plant-pathogenic fungus and unlikely to be pathogenic to wasps.

Bacteria

Bacteria are always present in the microflora of dead insects, regardless of their pathogenicity. It is therefore not possible to separate pathogenic species from saprophytic species without bioassay against healthy wasps. Among those species listed in Table

Table 1 Checklist of microflora and nematodes recorded in natural situations from wasp species of the genera *Vespula*, *Vespa*, and *Dolichovespula*.

Micro-organism/nematode	Host material	Site	Reference/source of reference
Fungi			
Alternaria sp.	*Vespula vulgaris* larva	England	Glare, Rose & Ratnieks unpubl. data
Alternaria tenus	*Vespula* sp. nest material	Colorado	Durrell 1965
Aspergillus sp.	*Vespula* spp. nest material	North America	MacDonald 1977
Aspergillus flavus	*V. vulgaris* adult	California	Gambino 1988
	V. vulgaris larvae	New Zealand	Glare et al. 1996
Aspergillus restrictus	*V. vulgaris* comb, larvae	California	Gambino & Thomas 1988
Aspergillus niveus	*V. vulgaris* comb, larvae	California	Gambino & Thomas 1988
Aspergillus niger	*V. vulgaris* comb, adult	California	Gambino & Thomas 1988
Aspergillus oryzae	*V. vulgaris* comb, larvae	California	Gambino & Thomas 1988
Aspergillus terreus	*V. vulgaris* comb	California	Gambino & Thomas 1988
Aspergillus wentii	*V. pensylvanica* larval gut	California	Gambino & Thomas 1988
Aureobasidium pullulans	*Vespula* sp. nest material	Colorado	Durrell 1965
Beauveria sp.	*V. pensylvanica* larvae	Hawaii	Nakahara (1980) *in* Gambino 1988
	V. vulgaris	Canada	Goettel *in* Humber 1992
Beauveria bassiana	*V. vulgaris* adult cadavers	California	Gambino & Thomas 1988
	Vespidae adult	Brazil	Daoust *in* Humber 1992
	Vespidae	Brazil	Lord & Magalhaes; Wraight *in* Humber 1992
	Vespula sp.	New Zealand	Barker et al. 1991
	V. germanica brood	New Zealand	Wigley & Dhana 1988
	"on a wasp"	Britain	Petch 1948
	Vespula maculata (L.)	North Dakota	Thomas & Poinar 1973
	V. germanica larva and adult	New Zealand	Landcare Research ICMP strains collection #8147 (1983) and #9271 (1986)
Beauveria brongniartii (=*B. tenella*)	*V. germanica*	New Zealand	Laird 1991
Beauveria "densa"	"on a wasp"	Britain	Petch 1948
Beauveria globuliferum (Speg.)	*Vespa* sp.	United States	Pettit (1895) *in* Petch 1924
Botrytis cinerea	"Nest paper" *V. germanica*	France	Acolat 1953
	"Nest paper"		Cymorek 1978
Cladosporium sp.	*V. vulgaris* comb	California	Gambino & Thomas 1988
Cordyceps ditmari	"a wasp"	Jura	Petch 1932b
	Vespa sp.	Europe	
	Vespa vulgaris	Europe	
	Vespa crabro	–	Weiser 1977
Cordyceps gentilis	"a wasp"	Sarawak	Petch 1932b
	"a hornet"	Thailand	
	Vespa velutia	Java	Kobayasi 1941
Cordyceps humberti (=*Hirsutella saussurei*)	*Vespa cinata*	Africa, Brazil, Indonesia, Malaysia	Petch, 1934; Kobayasi 1941
Cordyceps langloisii	*Vespa muraria*	North America	Kobayasi 1941
	"a wasp"	Jarva	Petch 1932b
Cordyceps smithii	*Vespa* adults	North America	Kobayasi 1941
Cordyceps sphecocephala	"wasp" adults	Thailand	Hywel-Jones 1995
	Vespa auraria	Japan, West Indies, Central America, South America, China	Kobayasi 1941
	"a hornet"	Britain	Petch 1932b
	compilation of early records	–	Cooke (1892) *in* Gambino 1988

(*continued overleaf*)

Table 1 (*continued*)

Micro-organism/nematode	Host material	Site	Reference/source of reference
	Dolichovespula media	–	Poelt & Jahn (1963) *in* Edwards 1980
Fusarium roseum	*Vespula* sp. nest material	Colorado	Durrell 1965
Hebeloma radicosum	*Vespula lewisii* decomposing nest material	Kyoto, Japan	Sagara et al. 1985
Hebeloma spoliatum	*V. lewisii* decomposing nest material	Kyoto, Japan	Sagara & Kobayashi 1979; Sagara et al. 1985
Hirsutella sp.	*V. germanica* worker	New Zealand	Wigley & Dhana 1988
Hirsutella saussurei (=*Cordyceps humberti*)	*Vespa cincta*	Ceylon	Petch 1924; Petch 1934
	Vespula sp.	New Zealand	Glare & Harris, unpubl. data
	"a wasp"	Panama	Petch 1948
	"a wasp"	Trinidad	
Hymenostilbe sp.	"wasp" adults	Thailand	Hywel-Jones 1995
Hymenostilbe sphecophila (=*Cordyceps sphecophila*)	wasp adult	–	Gray (1858) *in* Gambino 1988, Petch 1948
Isaria sp.	"wasp"	India	Thomas & Poinar 1973
Mucor varians	*Vespula* sp. nest material	Colorado	Durrell 1965
Paecilomyces farinosus	*V. pensylvanica* cadavers	California	Gambino & Thomas 1988
	V. germanica adult	New Zealand	Landcare Research ICMP strains collection #7819 (1982)
	Vespa crabro	Czechoslovakia	Kmitowa 1982
	V. sylvestris	–	Swanton (1919) *in* Samson 1974, Petch 1932a
Penicillium sp.	*Vespula* spp. nest material	North America	MacDonald 1977
Penicillium aurantio-candidum	*V. vulgaris* larval gut	California	Gambino 1988
Penicillium brevi- compactum	*V. vulgaris* comb	California	Gambino 1988
Penicillium chermesinum	*V. pensylvanica* larval gut	California	Gambino 1988
Penicillium citinum	*V. vulgaris* comb, larvae	California	Gambino 1988
Penicillium corylophilum	*V. pensylvanica* adult	California	Gambino & Thomas 1988
Penicillium decumbens	*V. pensylvanica* larval gut	California	Gambino & Thomas 1988
Penicillium humili	*V. pensylvanica* larval gut	California	Gambino & Thomas 1988
Penicillium lanosum	*V. pensylvanica* larval gut	California	Gambino & Thomas 1988
Penicillium phoenicium	*V. pensylvanica* larval gut	California	Gambino & Thomas 1988
Penicillium restrictum	*V. pensylvanica* larval gut	California	Gambino & Thomas 1988
Penicillium steckii	*V. vulgaris* comb, larvae *V. pensylvanica* larval gut	California	Gambino & Thomas 1988; Gambino 1988
Penicillium stoloniferum	*V. vulgaris* comb	California	Gambino & Thomas 1988
Penicillium thomii	*V. vulgaris* comb	California	Gambino & Thomas 1988
Phoma sp.	*Vespula* sp. nest material	Colorado	Durrell 1965
Scropulaiopsis brevicaulis	*V. vulgaris* comb	California	Gambino & Thomas 1988
Stemphylium ilicis	*Vespula* sp. nest material	Colorado	Durrell 1965
Bacteria			
Actinomycetes sp.	*V. vulgaris* larval gut *V. pensylvanica* larval gut	California	Gambino 1988
Bacillus cereus	*V. pensylvanica* larval gut	California	Gambino 1988
Bacillus stearothermophilus	*V. pensylvanica* larval gut	California	Gambino 1988
Micrococcus sp.	*V. pensylvanica* larval gut	California	Gambino 1988
Proteus sp.	*Vespa crabro germana* (Christ) larvae and pupae	Massachusetts	Steinhaus & Marsh 1962
Proteus vulgaris	*Vespa orientalis* larvae	Israel	Ishay et al. 1967
Serratia liquefacians	*V. pensylvanica* larval gut	California	Gambino 1988
Serratia marcescens	Guts of several social vespids, incl. *V. germanica*	–	Verge (1952) *in* Gambino 1988
	V. germanica	New Zealand	Glare et al. 1993

Micro-organism/nematode	Host material	Site	Reference/source of reference
Staphylococcus aureus	*Vespa orientalis* larvae	Israel	Ishay et al. 1967
Streptococcus sp.	*V. pensylvanica* larval gut *V. vulgaris* larval gut	California	Gambino 1988
Streptococcus-like form	*Vespa crabro germana* (Christ) larvae	Massachusetts	Steinhaus & Marsh 1962
Streptomyces avermitilis	*Vespula maculifrons* larvae	–	Parrish & Roberts 1984
Virus			
Cricket paralysis virus	*V. germanica* brood	New Zealand	Wigley & Dhana 1988
	V. vulgaris, Vespa crabro, D. media	–	Martignoni & Iwai 1986
Kashmir bee virus (Strain 1)	*V. germanica*	New Zealand	Wigley & Dhana 1988
Protozoa			
Unidentified microsporidian	*V. germanica* brood, adults *V. vulgaris* brood, adults	New Zealand	Wigley & Dhana 1988
Crithidia sp	*Vespula squamosa* worker intestines	USA	Urdaneta-Morales 1983
Nosema sp.	*Vespa* sp.	Canada	Van der Geest & Van der Laan 1971
N. apis	*Vespa* sp.	Japan	Van der Geest & Van der Laan 1971
N. sp.	*V. germanica* larvae	Australia	Anderson pers. comm. (Patent No. PCT/AU92/00289)
N. spp. (2 species)	*V. vulgaris* and *V. germanica* queens	New Zealand	P. Wigley & P. Scotti unpubl. data
Nematodes			
*"*Gordius vespae vulgaris*"	*V. vulgaris* abdomen	England	Baird (1853) *in* Poinar et al. 1976
* "a big worm"	*V. vulgaris* male abdomen	Austria	Kristof 1879
* "*Mermis pachysoma*"	*V. germanica*	England	von Linstow 1905
* "Nematode worms"	*V. vulgaris* adults	England	Beck 1937
* "Gordius" worms	*Vespula* sp. drones	England	Fox-Wilson 1946
* *Gordioidea* worms	*V. germanica* queens	Germany	Kloft 1951
* "Gorgiid worms"	*V. germanica* queens	Germany	Gauss 1970
Pheromermis pachysoma	*V. vulgaris* queen	England	Waterson & Baylis (1930) *in* Poinar et al. 1976
	V. germanica gynes, male, workers; *V. vulgaris* gynes; *D. sylvestris* gynes	England	Blackith & Stevenson 1958
	V. pensylvanica males and queen	California	Poinar et al. 1976
	D. arenaria	California	Poinar 1981
	Larvae, pupae and adults of: *D. saxonica*, *Vespa crabro* *V. vulgaris*, *V. germanica*	Austria	Kaiser 1987
Sphaerularia bombi	*V. rufa* and *V. vulgaris* queens	–	Bedding 1984
Steinernema carpocapsae	*V. germanica* queen cadaver	Australia	Bedding 1984
(= *Neoaplectana feltiae*)	*Vespula* sp.	Australia	Akhurst 1980

*All species later described as *Pheromermis pachysoma*.

1, *Bacillus* spp. and *Serratia marcescens* are often insect pathogens, but cannot be confirmed as such without confirming bioassays, as both genera contain both pathogens and saprophytes. Only isolates of *S. marcescens* and a *B. thuringiensis* β-exotoxin have had their pathogenicity or toxicity confirmed in bioassays. The bacterium *S. entomophila*, a pathogen of scarab beetles, proved non-pathogenic to *Vespula* spp. (Table 2).

Viruses

Viruses are not commonly associated with vespid death. Cricket paralysis virus has been recorded several times from *Vespula*, *Vespa*, and *Dolichovespula*, but there are no other records except for one Kashmir bee virus infection (Table 1). Wigley and Scotti (unpublished) used a *Drosphilla* cell line to detect virus replication, and found Cricket paralysis virus and Kashmir bee virus in two of 20 *V. ger-*

Table 2 Pathogen status of fungi, bacteria, and nematodes bioassayed against *Vespula* spp.

Agent	Host	Pathogenic*	Dose (cfu/wasp unless specified)	Reference
Fungi				
Aspergillus flavus	*V. vulgaris* larvae	Yes	175×10^3	Glare et al. 1996
	V. germanica larvae	Yes	262.5×10^3	
	V. vulgaris larvae	No	1.75×10^3	Harris unpubl. data
Beauveria bassiana	*V. vulgaris* larvae	Yes	9.05×10^3	Harris unpubl. data
		Yes	175×10^3	
		No	1.75×10^3	
Paecilomyces sp.	*V. vulgaris* larvae	No	175×10^3	Harris unpubl. data
Metarhizium sp.	*V. germanica* adults	Yes	?	Spradbery & Huppatz 1993
	V. germanica larvae	Yes	dust formulation	
Metarhizium anisopliae var. *anisopliae*	*V. vulgaris* larvae	Yes	12×10^3	Harris unpubl. data
		No	1.2×10^3	
Bacteria				
Serratia marcescens		No	5×10^6	Harcourt unpubl. data
S. entomophila		No	5×10^6	Harcourt unpubl. data
Avermectin B1 (from *Streptomyces avermitilis*)	*Vespula maculifrons* (Buysson) larvae	Yes	acute (38 ppm in food) chronic (3 ppm in food)	Parrish & Roberts 1984
Bacillis thuringiensis (11 varieties)	*V. vulgaris* workers	Yes	fed 1: 3 to 1:18 (50% sugar syrup :technical b-exotoxin)	Haragsim & Vankova 1973
B. thuringiensis Dipel (var. *kurstaki*)	*Vespula* larvae	Yes	High (4.84 to 9.1×10^3/larvae)	Gambino 1988
B. thuringiensis (var. *israelensis*)	*Vespula* larvae	Yes	Medium (1.7×10^2 to 4.31×10^4/ larvae)	Gambino 1988
B. thuringiensis Bactospeine (var *kurstaki*)	*Vespula* larvae	Yes	Low ($2.7 \times 10^1$to 9.7×10^5/larvae)	Gambino 1988
Nematodes				
Heterorhabditis bacteriophora	*V. pensylvanica* larvae	Yes	2×10^3 nematodes/ larvae	Gambino 1984
Steinernema (= *Neoaplectana*) sp. Hopland strain	*V. pensylvanica* larvae	Yes	2×10^3 nematodes/ larvae	Gambino 1984
Steinernema (=*Neoaplectana*) *carpocapsae* Italian strain	*V. pensylvanica* larvae	Yes	2×10^3 nematodes/ larvae	Gambino 1984
S. carpocapsae Leningrad strain	*V. pensylvanica*, *V. rufa atropilsa* and *Vespula* sp.	Yes	3000/ml feedingliquid	Poinar & Ennik 1972
S. carpocapsae Hopland strain	*Vespula* larvae	Yes	LD_{50} range: 8.6×10^1 to 5.66×10^2/larvae	Gambino 1988
Pheromermis vesparum	*V. vespula* and *V. germanica* colonies	Yes		Kaiser 1987
Steinernema feltiae	*V. pensylvanica*, *V. vulgaris* field colonies	Yes		Gambino et al. 1992
	V. germanica larvae	Yes	20–200/larvae	Guzman 1984
	V. germanica adults	Yes	200/adult	
	Vespula colonies	Yes	$2\text{–}100 \times 10^6$ nematodes/colony	Gambino 1988
	V. pensylvanica adults	Yes	19–34 desiccated nematodes/male+female	Wojcik & Georgis 1988
Protozoa				
Vairimorpha mesnili	*V. vulgaris* larvae	No	4.5×10^7 spores/larvae	Harris, Glare & Rose unpubl. data
Virus				
Kashmir bee virus (Strain 1)	*V. germanica* pupae	Yes	injected into host	P. Wigley & P. Scotti unpubl. data

Agent	Host	Pathogenic*	Dose (cfu/wasp unless specified)	Reference
Kashmir bee virus (Strain 2)	*V. germanica* pupae	Yes	injected into host	P. Wigley & P. Scotti unpubl. data
Cricket paralysis virus	*V. germanica* pupae	Yes	injected intohost	P. Wigley & P. Scotti unpubl. data
Sacbrood virus	*V. germanica* pupae	No	replicated within approx. 20% of hosts	P. Wigley & P. Scotti unpubl. data
Black queen cell virus	*V. germanica* pupae	No	replicated within approx. 20% of hosts	P. Wigley & P. Scotti unpubl. data

* Significant mortality compared to control.

manica nests surveyed. Kashmir bee virus strain 1 & 2 and Cricket paralysis virus injected into *V. germanica* pupae all resulted in 100% mortality (Wigley & Scotti, unpublished). Sac brood and black queen virus replicated in only some pupae (20%) after injection.

Protozoa

Protozoa are rarely pathogenic to wasps. However, members of the order Microsporida have occasionally been found attacking Vespinae (e.g., *Nosema* sp. (Patent No. PCT/AU92/00289)) was isolated from a diseased nest in Australia (D. Anderson pers. comm.). *Crithidia* is a genus in the class Kinetoplastida (trypanosomatids) which contains a number of entomogenous species, few of which are truly pathogenic.

Nematodes

Members of the genus *Steinernema* are generally pathogenic to insects, although there is only one naturally occurring infection on record from Vespinae (Akhurst 1980, listed a strain of "*Neoaplectana feltiae*" (=*Steinernema carpocapsae*) from *Vespula* sp. in Australia). Bedding (1984) found the cadaver of a single queen *V. germanica* containing many thousands of infective juvenile *S. carpocapsae*. *Steinernema* nematodes carry symbiotic bacteria, which are the cause of death. In trials using *S. carpocapsae* (Gambino et al. 1992), the nematode was effective against *Vespula* colonies (Table 2).

Sphaerularia bombi is usually a pathogen of bumble bees and Bedding (1984) suggests that a record of this nematode from queen wasps was the result of a cross infection, as the wasp nest was inside a bumble bee nest.

Mermithids are common pathogens of a number of insect groups and have complex life cycles. Two species, *Pheromermis pachysoma* and *P. vesparum*, which appear to be pathogenic to Vespinae, have been recorded in several countries (Tables 1 & 2). *Pheromermis pachysoma* infects paratenic hosts such as mosquitoes, caddis-flies and crane-flies, then encysts as a second stage nematode in various tissues, before infecting the final host, a vespid wasp that preys on the paratenic host (Poinar et al. 1976; Tanada & Kaya 1993).

DISCUSSION

Over eighty microbes have been isolated from wasps or wasp colonies, but few have been confirmed pathogenic by Koch's postulates. Several of the species recorded or confirmed as pathogenic may have potential as inundative biological controls (Table 3), but none could be used as as self sustaining classical biological controls. The challenge to use those with potential is to develop specific, effective delivery systems for use in the field to improve existing control.

Of the many fungi recorded, few could be expected to be pathogenic. Several bacteria are known, but are probably opportunistic pathogens or saprophytes. Mermithid nematodes are among the few common pathogens of Vespidae recognised that do achieve vertical transmission from colony to colony, but as their complex life cycle involves intermediate hosts, they would not be suitable for release as biological control agents in New Zealand because of potential non-target effects (Moller et al. 1991). There are few references to viruses and protozoa, which may partially reflect the lack of studies specifically looking for these groups in wasps. The list of microbial associates may overestimate the variety of diseases through misidentification: *Beauveria* species have been particularly difficult to separate, and some species recorded (e.g., *B. densa*, *B. globuliferum*) have been synomised with other species (e.g., de Hoog 1972; Mugnai et al. 1989).

Most records of disease are from lone adult wasps or vacated nests. Nest failure due to disease has seldom been recorded, but one such nest was reported in Australia, and a microsporidian (*Nosema* sp. (Patent No. PCT/AU92/00289)) was isolated from the nest (D. Anderson pers. comm.). This species has a broad host range and was likely to have entered the nest with infected prey.

The potential of a particular disease to control wasps will depend on its mode of infection and dispersal. Most fungi penetrate the insect cuticle directly, and can be spread between wasps by surface contact with a contaminated individual (T. Glare pers. comm.). Aerial spore dispersal between layers of comb could be impeded by the internal structure of horizontally oriented combs in wasp nests. Bacteria, viruses, and protozoa need to be ingested by adults or larvae. These diseases are most likely to be effective where they are spread quickly through trophallaxis, as once obvious symptoms develop, diseased individuals are likely to be removed from a nest before they can infect others. If disease does enter and spread throughout a nest, the mortality of a large number of colony members may not be sufficient to ensure total destruction of the nest, as wasps increase worker reproduction to compensate. Also, capped cells may provide refugia from pathogens for developing pupae.

As wasps are not innately resistant to generalist insect pathogens such as *Beauveria*, *Metarhizium* and *Steinernema*, we must seek other reasons for the lack of recorded diseases of social wasps. Colonies may have group-level defences not available to individuals. Abandoned wasp nests and colonies in late-season decline reveal an abundance of fungi, primarily *Aspergillus* and *Penicillium* spp. (MacDonald 1977). Wasps may actively suppress such growth in the mid-season through hygienic behaviour and/or chemical defence. Workers may produce and apply antimicrobial compounds to the nest walls, or gather a fungicide during fibre collection to incorporate into the wood pulp (MacDonald 1977). Conjugated noradrenaline, which is produced in the saliva of Hymenoptera, has been isolated from the nest envelopes of *Vespula* sp. (Bourdon et al. 1975; Lecomte et al. 1976) and may act as an antimicrobial compound (Akre & MacDonald 1986). The larval saliva of *Vespula* sp. has an antimicrobial effect (Gambino 1993). Wasp venom may also have antimicrobial activity, as in ants (Jouvenaz et al. 1972).

Worker removal, or the insulation of dead and diseased nestmates from active areas of the colony, may restrict establishment and spread of microbial pathogens. Cadaver removal from the nest vicinity

Table 3 A summary of isolates common to or demonstrated to be pathogenic or toxic to wasps from Tables 1 and 2.

Pathogen	Comments
Fungi	
Aspergillus flavus	Often found in nest material and shown to be pathogenic
Beauveria spp.	Shown to be pathogenic, naturally occurring pathogen often isolated from wasps
Metarhizium anisopliae	Generalist pathogen never associated naturally with wasps but virulent in bioassay
Cordyceps, Hirsutella and *Hymenostilbe* spp.	Naturally occurring pathogens but difficult to culture and apply
Bacteria	
Bacillus thuringiensis	β-exotoxin shown to be toxic but no specificity
Protozoa	
Nosema spp.	Some potential but few experimental infections
Nematodes	
Steinernema and *Heterorhabditis* spp.	Culturable, generalist pathogens shown to be virulent, may be difficult to infect colonies
Pheromermis spp.	Complicated life cycle makes use difficult
Virus	
Kashmir bee virus (Strain 1&2) and Cricket paralysis virus	Shown to kill wasps, but no methods known for infecting host in field

by workers also reduces the likelihood of researchers discovering diseases during nest dissections. Of cadavers found outside nests, those infected by fungi are most likely to be visible, due to mycelial growth and sporulation. The lack of known viral and protozoan diseases may be related to the fact that individuals infected with these show few obvious outward signs of disease and would easily be overlooked unless these diseases were specifically being looked for.

In the future, the discovery of more diseases of wasps, capable of infecting entire nests or being carried by queens, may increase our options for using pathogens against social wasps. Such diseases are less likely to be found in areas of relatively recent introductions, such as New Zealand, but systematic searching throughout the recorded range of social wasps may yet yield new records.

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